

ML Specification Report — playground-series-s5e1

Forecasting Sticker Sales · our from-scratch agent (level/shape structural Ridge + honest-gated tree-search v3)

Figures grounded in the competition's STATUS.md, config.yaml, dossier.json, experiments.json, experiments_tree_run2.json, final_selection.json, injection_ledger.json, and the join logs, with leaderboard figures from the study's frozen three-way benchmark table (FIRING_CLASS_THREE_WAY.md).

Overview

The task is to forecast daily sticker sales (`num_sold`) for 90 retail series — 6 countries $\times$ 3 stores $\times$ 5 products — across all of 2017–2019, trained on 2010–2016 history (metric: MAPE, minimize). We solve it with the full v5 pipeline: a dossier-primed run with external-data injection (World Bank GDP per capita, national holiday calendars), whose decisive move is **level/shape decoupling** — a country-year *level* anchored on GDP-per-capita times a convergence-ratio rule, and a structural Ridge *shape* model fit on `log(y / level)`. Our champion is the **solo structural Ridge** `ridge_level_conv50`, scoring **CV MAPE 6.9281** (pooled rounded over expanding-year folds; raw-clip1 6.9719). A 60-node tree search found **no honest improvement** — every fitted blend collapsed under leave-year-out reweighting — so the pre-registered linear champion was submitted. On the frozen three-way benchmark our submission is the **best private-leaderboard score of the three agents** (private MAPE 0.12752, percentile 76.5, vs NVIDIA 0.14594 / AIDE 0.17917), with the paired verdict confirming mine > AIDE and the other pairs statistically indistinguishable.

Why it matters. Multi-country retail demand forecasting drives inventory, logistics, and revenue planning; as a three-year future-window extrapolation it also tests whether an agent can recognize that the cross-country level must be carried by an external macroeconomic covariate rather than extrapolated by a tree model.

Data

Purpose of Data. Forecast daily units sold per (date $\times$ country $\times$ store $\times$ product) series — a tabular time-series regression Playground Series episode, target `num_sold`. **Data Format** is clean **tabular CSV** with four raw predictive columns (`date`, `country`, `store`, `product`) once `id` and the target are removed. **Data Volume** is **230,130 training rows (2010-01-01 $\rightarrow$ 2016-12-31, 2,557 days) / 98,550 test rows (2017-01-01 $\rightarrow$ 2019-12-31, 1,095 days)**, a complete 90-series daily panel on both sides.

Data Quality has one structural wrinkle: **3.85% of train targets are null**, and EDA shows this is **censoring of low values, not missing-at-random** — Kenya's cheapest series has non-null min = p10 = median = 5 with the null rate falling monotonically as demand grows (0.94 in 2010 $\rightarrow$ 0.08 in 2016), concentrated on weekdays; a Canada series is analogously censored at $\sim$ 200; 2 series are all-null. Policy: nulls are dropped

from fitting and excluded from CV scoring (MAPE is undefined against NaN). The strongest EDA signals, all pointing at a structured multiplicative panel, were:

- **Log-additive structure:** additive main effects on `log(y)` reach $R^2 = 0.979$, rising to **0.989** with a country $\times$ product interaction — strongly structured synthetic data (the tpsjan22 regime, not s3e19).
- **Store share is constant** (std 0.00035); **product share follows sinusoids** with 104-week and 52-week periods — bounded Fourier features, no extrapolation hazard.
- **GDP proportionality is converging:** the cross-country dispersion of `total(country, year) / gdp_pc` collapses **0.185 (2010) $\rightarrow$ 0.049 (2016)**, with every country's ratio trending toward ~ 1.0 — the regime the level anchor exploits.
- **A 2015 regime shift:** yearly totals fell **−10.9% in 2015 while GDP grew**; fold-2015 is the hardest year for every model.
- **Seasonality:** dow multipliers Mon–Thu 0.947 / Fri 1.00 / Sat 1.05 / Sun 1.157 (two country groups by dow profile), a Christmas/New-Year ramp to $1.30\times$ around Dec 25–Jan 5, and a minor mid-May bump (doy 127).

Annotation Guidelines. The label is `num_sold`, an integer daily count; submissions are **continuous forecasts** scored by MAPE — a relative-error regression metric, so the multiplicative (log-space) treatment and denominator-side protection matter more than raw squared error.

Feature Set. From the four raw columns plus two external joins the pipeline builds `features.parquet` (26 features for the GBDT arm) and a wider explicit ridge design:

Group	Features
Categoricals	<code>country_code</code> , <code>store_code</code> , <code>product_code</code>
Calendar / cyclical	<code>dow</code> , <code>doy</code> , <code>month</code> , <code>doy_sin1..4</code> / <code>doy_cos1..4</code> , <code>c104_sin1..2</code> / <code>c104_cos1..2</code> , <code>c52_sin1..2</code> / <code>c52_cos1..2</code>
External joins	<code>gdp_pc</code> / <code>log_gdp_pc</code> (World Bank per-capita, <code>merge_year_safe</code> lag = 0, snapshot 2026-07-13); <code>is_holiday</code> , <code>days_to_holiday</code> , <code>days_from_holiday</code> (holidays pkg 0.101, 6 countries, 10,530 holiday rows)
Ridge design (champion)	country/store/product main effects + $c \times p$ + $s \times p$ + $c \times \text{dow}$ interactions, Fourier (doy 8, product $\times 104w$ 2, product $\times 52w$ 2), per-holiday-name $\times$ offset(−5..+10) dummies

Both joins are leak-audited: the GDP join matched **328,680 / 328,680 rows** (0 unmatched, leakage check passed); holidays are a deterministic future calendar (no leakage possible).

Splitting strategy. The dossier pinned this before EDA: the test window is a pure **three-year future** (2017–2019 after 2010–2016), so validation is **expanding-year folds** — **train < y, validate == y, for y $\in$ {2014, 2015, 2016}** — and shuffled KFold is *forbidden* (a $4.5\times$ optimism bias was measured for it on s3e19). The scheme is deterministic (no fold seed needed), and the pooled score is computed with the submission post-processing (clip + round-to-int) *inside* the metric so every search decision optimizes the submitted quantity. Fold-2015, the regime-shift year, dominates the pooled score for every model (10.3–11.9 vs ~ 5.1 – 5.5 for 2014/2016), and the honest gates below turn on exactly that fold. All in-run decisions were **CV-only** (no leaderboard feedback during the run); the finished submission was then scored once on the frozen three-way benchmark.

Models & Architecture

Purpose of Architecture. A daily-sales forecaster that minimizes MAPE under future-window extrapolation. **Architecture Type** is — unusually for this report set — **not a GBDT blend but a solo structural linear model**: a two-part **level/shape decomposition**. The *level* `level(country, year)` is an anchor equal to `gdp_pc × conv50 ratio` — each country’s sales-to-GDP-per-capita ratio carried forward under the `conv50` rule, i.e. 50% continued convergence toward the common cross-country ratio (the convergence fraction sweep 25/75/100 scored 7.258 / 7.542 / 8.404 against `conv50`’s family best, a clean U-shape). The *shape* is a **Ridge regression ($\alpha = 1.0$) on $\log(y / \text{level})$** over the explicit design above. A LightGBM pool (log-target, ratio-target, MAPE-weighted, and level-anchor variants) was raced throughout and populated the blend pools, but every blend was rejected by the honest gate.

Each model takes the same **Input Format** — a numeric feature row per (date × series) — of **Input Dimension 26 features** in the GBDT arm; the ridge design width is not separately recorded. **Model Complexity** for the champion is that of a closed-form linear solve (one coefficient per design column) — no trees, no iterative training; parameter count not recorded.

Training procedures

Training proceeds as a **baseline race** → **level/shape iteration** → **blend gate** → **tree search** → **honest final selection** ladder, all on the identical expanding-year folds:

Stage	Configuration	pooled MAPE
3 — baselines (exp #1–5)	best: <code>lgb_join</code> (26 features, log target, GDP join); also <code>lgb_ratio</code> 11.808, <code>lgb_mapew</code> 11.085, <code>ridge_struct</code> 13.675, <code>ridge_cyc</code> 11.086; naive 12.228	9.195
iter — level/shape decoupling (exp #6–9)	LGB <code>shape</code> on <code>log(y/level)</code> , <code>level</code> rules last/avg2/trend/conv → 8.061 / 10.007 / 7.955 / 7.099	7.099
iter — conv sweep + ridge shape (exp #11–15)	<code>conv25/75/100</code> 7.258/7.542/8.404; <code>ridge_level_conv50</code> raw 6.9719 → rounded 6.9281	6.9281
blend (exp #16)	greedy Caruana 3-member blend — in-sample 6.858, honest leave-year-out 7.619 → REJECTED	(rejected)
tree search v3 (exp #17)	60 nodes: in-sample best 6.764 (38-member mega-blend) → no honest improvement	(rejected)
final honest selection (exp #18)	root solo + clip ≥ 5 + round-to-int	6.9281

The **Loss Function** is L2 on the log-space shape target (`log(y/level)`), with model selection on rounded pooled MAPE; the metric-mimicking alternative — L1 with 1/y sample weights (`lgb_mapew`) — was raced and lost badly (11.085 vs 9.195 for plain L2-on-log), settling the dossier’s open objective question by CV. The **Optimization Algorithm** for the champion is a closed-form ridge solve ($\alpha = 1.0$) — **not** SGD/ADAM and not even boosting; the LGB pool members use histogram gradient boosting. There is **no Learning Rate**

for the champion (*N/A* — *closed-form linear solve*), **no Learning Rate Scheduler**, and **no Batch Size** (*N/A* — *full-dataset fitting, not mini-batched*).

Training Duration: the tree-search stage ran **945.8 s of wall-clock over 60 evaluated nodes** (48 solo / 12 blend; stop = hard 60-node budget cap); the root champion refits in **9.2 s**. Runs were foreground, seeds fixed, threads = 10. **Training Memory** was not explicitly capped — a 230K-row panel is trivial on the single arm64 CPU machine, peak **not recorded**. **Transfer Learning** is *N/A* (*no pretrained weights*); its analogue here is the **v5 dossier priming** — four matched task priors (TASK-TS-FUTURE, TASK-TS-CALENDAR, TASK-MAE-METRIC, TASK-BLEND-DECOR) injected the split policy, the GDP ratio/join race, the holiday windows, and the postprocess-inside-metric rule before any model was fit (injection ledger: 8 proposed / 3 realized data-layer / 5 consumed configs / 0 unrealized). **Data Augmentation** is *N/A* (*tabular*); the analogues are the two external-data joins and the Fourier harmonics.

Reproducibility Standards are strict: fixed seeds throughout, a fresh tree-search evaluator + cache built for this run (lane isolation — the archived s5e1 caches and evaluators from the arms study were never touched, and `experience.md` contained no self-citing bullets to exclude), and the search root **digit-verified at 6.928121** against the linear stage before searching. The search's honest bookkeeping is fully logged: evals-to-match-linear-best = 1 (the root itself), 0 dedup rejections, 0 cost-guard firings, the burst sanity gate fired 2× (EXPL_CTRYTREND 7.520, EXPL_MAPEW 11.054 — both plateaued at seed), and 17 backtrack entries.

Inference procedures

Decision Threshold is *N/A* — MAPE is a regression forecast metric, so we submit continuous predictions and never threshold. Post-processing is **clip** ≥ 5 (the train minimum; a denominator-side MAPE protection, OOF 6.9279 vs 6.9281 for clip ≥ 1) then **round-to-int** (`num_sold` is a count; rounding is worth 6.9719 $\rightarrow$ 6.9281 on the champion and was applied inside the metric for all decisions). `submission.csv` is the champion's full-train test prediction — all 98,550 rows. **Inference Duration** was not separately profiled (a linear predict over 98,550 rows is sub-second) and **Inference Memory** is negligible (not capped).

Evaluation & Benchmarking

Task for Performance Evaluation. Forecast 2017–2019 daily sticker sales for all 90 series. **Performance Metrics.** MAPE (minimize) is the sole competition metric; our champion reaches **CV MAPE 6.9281** (pooled rounded, expanding-year folds; raw-clip1 6.9719, whose per-fold scores are 5.104555 / 10.294318 / 5.533216). The decisive quality control is the **honest gate**: the greedy blend's in-sample 6.858 collapsed to 7.619 under leave-year-out reweighting, and the final LOFO audit ranked every alternative behind the pre-registered root — root_solo **6.9281** / eqw_top3 7.1170 / best_solo_lofo 7.3070 / blend_refit 7.4875 / mega_refit 7.5753. The tree search's in-sample 6.764 was therefore rejected; its 15.8 minutes bought a *negative* result — the third competition in a row confirming that in a structure-dominated task with a homogeneous pool, fitted-weight blends overfit the few validation folds.

Performance Benchmarking. Between-agent claims use the **frozen leaderboard numbers**; the local CV above describes the pipeline's internal view only (and is quoted in percentage points, whereas Kaggle reports

MAPE as a fraction — different unit conventions, not directly comparable digits). All three agents submitted to the real leaderboard, joining the frozen baseline 2026-07-31:

Agent	Public MAPE	Private MAPE	Percentile rank
Our agent (mine)	0.10236	0.12752	76.5
NVIDIA	0.05726	0.14594	69.3
AIDE	0.12573	0.17917	48.1

Our agent takes the private leaderboard (`lb_winner` = mine on private; paired verdict **M > A**, the other pairs statistically indistinguishable). The shape of the result is instructive: NVIDIA’s reproduction posts a far better *public* score (0.05726) but falls to 0.14594 on *private* — a public → private shake-up — while our CV-only, honest-gated submission degrades least and finishes at percentile 76.5, the best final standing of the three. The caveat is stated plainly: only the mine-vs-AIDE gap survives the paired significance test, so the headline claim vs NVIDIA is “ahead on private, not statistically separable”, not a proven win. This run used the full v5 pipeline — dossier priming plus external-data injection with the knowledge library pinned; the dossier’s unresolved join-vs-ratio form question (LB historically picked `join_feature`, CV picks the level/ratio form) stays open here, since the level-anchor form’s CV margin was large (6.93 vs 9.16 rounded) and it is what was submitted.

Fields marked “not recorded”: ridge design width / parameter count; training peak memory; inference duration and memory; per-stage wall-clock outside the tree search (only the 945.8 s search and the 9.2 s root refit are logged).